Central Operations Case Study Prompt

Prompt: *This case study is a series of discrete projects to set up / fix RCM processes that are stopping the normal process from delivering revenue to our clients.*

You only need a high level understanding of the broader Revenue Cycle Management process, but you will be expected to conduct some research to complete each specific problem listed below.

It is the process of collecting clinical information, submitting claims to insurance companies, and collecting + reporting revenue back to our customer, the healthcare practice.
You can treat it as an abstract, simplified manufacturing process with the following (rough) steps:

1. ***Extractions:*** We pull the data from our client's database into ours. *99% automated.*
2. ***Submissions:*** We reformat and structure (and often tweak specific fields with automated rules). Some of these fail filters and need to be worked by hand, but for normal claims these are automated. These are submitted to specific insurance companies to be processed. *99% automated.*
3. ***Reconciliation:*** We find out what the insurance company decided. Did we even send to the right insurance company? Were these claims they paid or denied, how much, and why was this the decision? These decisions either come to us via automated set ups, or are found via our offshore call center. *85% automated*
4. ***Denials:*** We review and determine how a claim was submitted wrong. This could be something complicated like how it was coded, or easy like we put the wrong zip code for a patient. Currently is the most manual process, though often we can work 100 that are denied for the same reason in bulk. *0% automated*

Format:
Please present in short memo or presentation deck format, leveraging flow-charts, spreadsheets, diagrams, or whatever else needed to convey your point.

The end-goal is to present an operational plan to own and dramatically improve each given metric that is rooted in numbers, immediately implementable, and able to iterate based on early findings.

If anything is vague or unclear, please research RCM and for specific data, please make and note your assumptions.

Across all these problems, your resources are the following (though using less is better):

1. Three Mountain View Data Entry / Calling Operators
2. Twenty India Team Data Entry / Calling Operators
3. Two Hours of a single engineer's time per week

4. Chat GPT, Google, calling insurance companies themselves and asking, etc.

Problem 1: Workers Compensation Claims

Workers Compensation claims work differently from regular claims because they require being physically mailed.

We have an integration where we can submit PDFs via an internal tool where an outside vendor will do the physical mailing for us, however pulling and formatting these PDFs is still a manual tool run by our offshore team.

However, our offshore team is finding they are getting stuck on 90% of claims. There should be 500 per day but they are only able to complete 50.

If you had a tool we built fully in house, that an India offshore team was using to submit claims, but it was failing 90%+ of the time, how would you:

1. Diagnose why it is failing so much?
2. If the problem is our tooling, how do you solve?
3. If the problem is our offshore team's workflows, how do you solve?
4. If there is something else going wrong, what could it possibly be?
5. What key metrics would you track afterwards to make sure it is functioning?

Problem 2: Closed Encounters Analysis

Background:

Ops Case Study Dataset - Sample DB Data.csv

Ops Case Study Dataset - Sample EHR Data.csv

You have been provided with two datasets (linked above) in CSV format:

1. **Closed Encounters from the Client's EHR**: A list of encounters that have been closed in the client's electronic health record (EHR).
2. **Imported Closed Encounters**: A list of closed encounters that have been successfully imported into Normandy.

Each dataset contains information about Patient Name, Date of Service, and Rendering Provider, which together form a unique id for an encounter, alongside procedure-level details.

Some encounters in the client's EHR may not be present in our database, and your task is to identify these missing encounters and come to a conclusion as to why the encounters weren't imported.

Instructions:

1. **Task 1: Identifying Missing Encounters**
    - Compare the two datasets and identify which encounters from the **Client EHR Closed Encounters** file are not found in the **Imported Closed Encounters** file.
    - Provide a list of these missing encounters in the form of the unique ID mentioned above (Patient Name, Date of Service, and Rendering Provider)
2. **Task 2: Investigating the Cause**
    - Analyze the missing encounters and identify any patterns or reasons why they might not have been imported into our database.
    - Summarize your findings and explain why certain encounters were not imported.

Hints:

- Identify trends within the imported encounters vs the encounters that weren't imported
- You can use any tools you're comfortable with (e.g., Excel, SQL, Python) to analyze the data and answer these questions.

Deliverables:

- A list of missing encounters and their associated details.
- Some analysis into the associated factors with these missing encounters, with the goal of identifying potential reasons for why these encounters weren't imported into our database.

Problem 3: Reconciliation - Claims Not Found

One problem we found is that a lot of claims end up in “claim not found.” We submit them to insurance companies and then never hear back.

When our reconciliation process runs, our team attempts to call each insurance company and normally gets the status, however sometimes they cannot find the claim.

If there were 10,000 of these claims across 150 insurances, how would you close out these claims and set up a process so we don’t run into this kind of backlog again? Focus on how you would:

1. Bucket the claims / prioritize to make the most use of you and your team’s time
2. Diagnose why they are missing (or if why they are not missing, why we couldn't find them earlier)
3. What are the likely actions we would take to resolve these claims?

Problem 4: Debugging Revenue

This is a 3 part problem. Please answer all parts. Be as specific as possible, show all intermediate work, and include links to helpful resources.

Part A

For CPT Code 97110, Medicare pays physicians $38.73 in MAC Locality 0210201, but pays physicians $27.55 in MAC locality 0730200.

Please explain this discrepancy and how Medicare determines physician payment per CPT code.

Be as specific as possible, show all intermediate work, and include links to helpful resources.

Part B

We submitted claim #1 to Aetna (insurance company) with the following CPT codes for one of our clients: 97110, 97140, G0283, 97112; all these procedures were performed on the same day for patient #1.

In the claim #1's remittance, Aetna decided to pay out $20 for CPT code 97140. However, when we submitted claim #2 to Aetna with just 97140 for patient #2, Aetna paid out nearly $40.

Why is there a difference in the payout for 97140 between claim #1 and claim #2? You can assume both patients have the same type of insurance policy and that Aetna did not make an error.

Be as specific as possible, show all intermediate work, and include links to helpful resources.

Part C

A Physical Therapist provided the following details regarding a patient's visit and would like us to submit a claim for this patient's visit to Medicare.

Timed Minutes	CPT Code	Units Billed
38	97110	2
24	97112	1
45	G0283	3

Please explain any issues with this and provide a way to optimize dollars paid for this claim.

Be as specific as possible, show all intermediate work, and include links to helpful resources.